Why Measure Theory

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1 Introduction: Riemann Integration

The Riemann integral works well for continuous functions, but it behaves badly under limits: a sequence of Riemann integrable functions can converge pointwise to a function that is not Riemann integrable at all. The goal of this paper is to show exactly how this failure happens, to explain it in the language of measure, to show why we cannot simply measure every subset of $\mathbb{R}$, and finally to show that integrating with respect to a measure resolves the original example.

1.1 Definitions

I will begin with the definitions of partitions, and upper and lower sums.

Definition 1 (Partition)**.** Let $[a, b] \subset \mathbb{R}$ then P, a partition, is defined as

$$P = \{x_0, x_1, \ldots, x_n\}$$

Such that

$$a = x_0 < x_1 < \ldots < x_n = b$$

Note. P is a finite collection of points that subdivides the interval [a, b].

Importantly it is finite and partitions the domain of a function f on $\mathbb{R}$.

Definition 2 (Upper sums)**.** Let f be a bounded function $f\colon [a,b] \to \mathbb{R}$, the upper sums of f for a partition P are defined as

$$U(f,P) = \sum_{k=1}^{n} (x_k - x_{k-1}) \sup_{[x_{k-1},x_k]} f$$

Where $U(f)$ is defined as

$$U(f) = \inf_{P} U(f,P)$$

Definition 3 (Lower sums)**.** Let f be a bounded function $f\colon [a,b] \to \mathbb{R}$, the lower sums of f for a partition P are defined as

$$L(f,P) = \sum_{k=1}^{n} (x_k - x_{k-1}) \inf_{[x_{k-1},x_k]} f$$

Where $L(f)$ is defined as

$$L(f) = \sup_{P} L(f,P)$$

Now defining Riemann Integration.

Definition 4 (Riemann Integration)**.** Let f be a bounded function $f\colon [a,b] \to \mathbb{R}$.

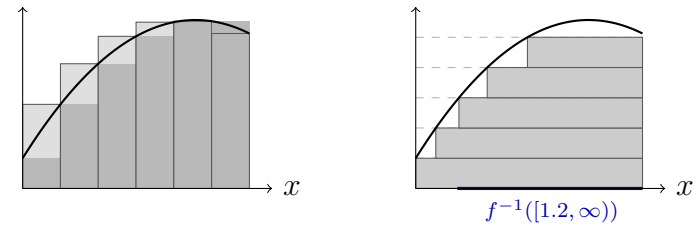
If for every $\epsilon > 0$ there exists a partition P_ϵ such that

$$U(f,P_\epsilon) - L(f,P_\epsilon) < \epsilon \tag{1}$$

then $f(x)$ is Riemann Integrable with

$$\int f\,dx = U(f) = L(f) \tag{2}$$

Notice from the definition that a function being Riemann Integrable relies on us finding partitions on which the upper and lower sums become arbitrarily close. In general, how can we tell in advance whether a function is Riemann Integrable?



Riemann: partition the domain

Lebesgue: partition the range

Figure 1: Riemann sums cut the domain into intervals; the Lebesgue integral groups points by the value of f and asks only for the *measure* of each such set.

1.2 Riemann Failure

1.2.1 Dirichlet Function

Let the sequence of numbers $\{r_1, r_2, \ldots\}$ enumerate the rationals on $[0, 1]$.

Let $f_n(x)$ be a series of functions defined on $[0, 1]$ of the form

$$f_n(x) = \begin{cases} 1: & x \in \{r_1, r_2, \ldots, r_n\} \\ 0: & x \notin \{r_1, r_2, \ldots, r_n\} \end{cases} \tag{3}$$

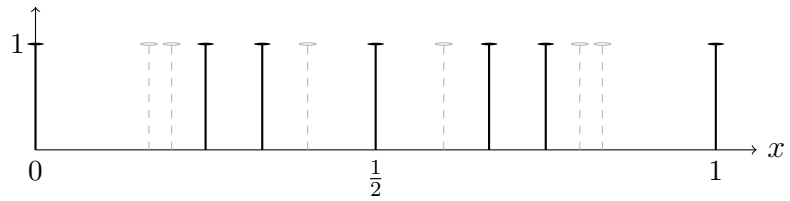


Figure 2: f_7 for the enumeration $0, 1, \frac{1}{2}, \frac{1}{3}, \frac{2}{3}, \frac{1}{4}, \frac{3}{4}, \ldots$: finitely many spikes (solid), each invisible to the integral. Dashed spikes are switched on for larger n; in the limit every rational has one.

1.2.2 Riemann Integration

I want to show each f_n is Riemann integrable. From the definition I must show

$$U(f_n, P_n) - L(f_n, P_n) < \epsilon$$

for every $\epsilon > 0$, for some partition P_n.

To begin fix $n \in \mathbb{N}$.

Take the partition P_n such that $(x_k - x_{k-1}) \leq \frac{\epsilon}{4n}$ for all consecutive pairs of points in P_n (the last interval may be shorter).

$$P = \{0, \frac{\epsilon}{4n}, \frac{2\epsilon}{4n}, \ldots, 1\}$$

The total number of intervals in which there may exist rational numbers is 2n; since there are n rationals where each can be in at most 2 intervals at once.

Note. if $r_k = x_k$ for instance, then $[x_{k-1}, x_k]$ and $[x_k, x_{k+1}]$ both contains r_k

From the definition of upper sums

$$U(f_n, P_n) = \sum_{k=1}^{n} (x_k - x_{k-1}) \sup_{[x_{k-1}, x_k]} f$$

The length of each interval as defined above is $(x_k - x_{k-1}) = (\frac{\epsilon}{4n})$, there exists at most $2n$ of these intervals, each with $\sup_{[x_k - x_{k-1}]} f = 1$.

All other possible intervals have a max combined length of 1, each with $\sup_{[x_k - x_{k-1}]} f = 0$.

This gives

$$U(f_n, P_n) \leq (\frac{\epsilon}{4n})(2n)(1) + (1)(0) = \frac{\epsilon}{2}$$

For the lower sums, since $\mathbb{I}$ is dense in $\mathbb{R}$, between any two real numbers there exists an irrational.

Meaning on any interval $[x_{k-1}, x_k]$, $\inf_{[x_{k-1}, x_k]} f = 0$.

This gives

$$L(f_n, P_n) = 0$$

This gives, for our fixed n and any $\epsilon > 0$,

$$U(f_n, P_n) - L(f_n, P_n) \leq \frac{\epsilon}{2} < \epsilon$$

meaning f_n is Riemann integrable. Moreover $0 = L(f_n) = U(f_n) \leq \frac{\epsilon}{2}$ for every $\epsilon > 0$, so

$$\int_0^1 f_n \, dx = 0$$

1.2.3 Limit

Letting $n \to \infty$ gives the limit of f_n as

$$\lim_{n \to \infty} f_n = \begin{cases} 1: & x \in \mathbb{Q} \cap [0,1] \\ 0: & x \notin \mathbb{Q} \cap [0,1] \end{cases} \tag{4}$$

Call this function f.

Now both $\mathbb{Q}$ dense in $\mathbb{R}$ as well as $\mathbb{I}$ dense in $\mathbb{R}$.

Meaning for any partition P, on any interval $[x_{k-1}, x_k] \subset [0,1]$ there exists both a rational and irrational.

This gives $\inf_{[x_{k-1},x_k]} f = 0$ and $\sup_{[x_k - x_{k-1}]} f = 1$ on every interval.

Which forces

$$U(f,P) = 1$$

and

$$L(f,P) = 0$$

for every partition P. Implying $U(f) = 1$ and $L(f) = 0$ which says that f is not Riemann integrable.

1.3 Lebesgue Explanation

1.3.1 Measure 0

To understand measure 0 first consider the concept of length.

Definition 5 (Length)**.** The length of an open interval $(a,b) \subset \mathbb{R}$ is defined as

$$l((a,b)) = b - a \tag{5}$$

Now to define measure 0.

Definition 6 (Measure 0)**.** A is a measure zero set if for every $\epsilon > 0$ there exists a countable covering $\bigcup_n O_n$ where each O_n is an open interval such that

$$A \subset \bigcup_n O_n \tag{6}$$

and

$$\sum_n l(O_n) \leq \epsilon \tag{7}$$

1.3.2 Lebesgue Criterion

By the Lebesgue criterion

Theorem 1 (Lebesgue Criterion)**.** *A bounded function $f\colon [a,b] \to \mathbb{R}$ is Riemann integrable iff the set of discontinuities is measure 0*

We have now defined the class of Riemann integrable functions in terms of a concept of measure which did not previously exist in the context of Riemann integration.

1.3.3 Dirichlet Discontinuous Everywhere

Recall that

$$f = \lim_{n\to\infty} f_n = \begin{cases} 1: & x \in \mathbb{Q} \cap [0,1] \\ 0: & x \notin \mathbb{Q} \cap [0,1] \end{cases}$$

I want to show that f is discontinuous on all of $[0,1]$.

First observe the definition of continuity.

Definition 7. A function f is continuous at a point c if for every $\epsilon > 0$ there exists a $\delta > 0$ such that whenever $|x - c| < \delta$ it follows that $|f(x) - f(c)| < \epsilon$

f Discontinuous

Begin by fixing $\epsilon \leq 1$, I will take $\epsilon_0 = \frac{1}{2}$. Choose any point $c \in [0,1]$.

Again both $\mathbb{Q}$ and $\mathbb{I}$ dense in $\mathbb{R}$.

First consider $c \in \mathbb{Q} \cap [0,1]$ then for any δ chosen there exists some irrational point

$$x \in (c-\delta, c+\delta) \implies |f(x) - f(c)| = 1 \geq \epsilon_0 \tag{8}$$

Consider the case $c \in \mathbb{I} \cap [0,1]$ then for any δ chosen there exists some rational point

$$x \in (c-\delta, c+\delta) \implies |f(x) - f(c)| = 1 \geq \epsilon_0 \tag{9}$$

Meaning the set of discontinuous points of f is all of $(\mathbb{Q} \cap [0,1]) \cup (\mathbb{I} \cap [0,1]) = [0,1]$ where $|[0,1]| = 1$.

Note. I do not prove it here but it is proved in section 2.1

fn Measure 0 Discontinuities

To show that the set of discontinuities for every f_n is measure 0 first consider the set of points at which f_n is discontinuous, call it D_n.

For any points $c \in \{r_1, r_2, \ldots, r_n\}$ since $\mathbb{I}$ dense in $\mathbb{R}$ we have for a fixed $\epsilon < 1$ and any $\delta > 0$

$$|x - c| < \delta \implies |f(x) - f(c)| = 1 \geq \epsilon$$

Making

$$D_n = \{r_1, r_2, \ldots, r_n\}$$

For any points $c \notin \{r_1, r_2, \ldots, r_n\}$ we can take $\delta = \frac{\min(|c-r_1|, |c-r_2|, \ldots, |c-r_n|)}{2} > 0$ since there are a fixed amount of rationals for any fixed $n \in \mathbb{N}$.

This would give us for

$$|x - c| < \delta \implies |f(x) - f(c)| = 0 < \epsilon$$

making f_n continuous on any points $x \notin \{r_1, r_2, \ldots, r_n\}$.

Now consider the open covering $\bigcup_k O_k$ such that each

$$O_k = (r_k - \frac{\epsilon}{2^{k+1}}, r_k + \frac{\epsilon}{2^{k+1}})$$

Then

$$\begin{aligned} \sum_k l(O_k) &= \sum_k \frac{\epsilon}{2^k} \\ &= \frac{\frac{\epsilon}{2}}{1 - \frac{1}{2}} \\ &= \epsilon \end{aligned}$$

giving

$$\sum_k l(O_k) \leq \epsilon$$

making D_n measure 0 by definition. The same covering, taken over the whole enumeration $r_1, r_2, \ldots$, shows that $\mathbb{Q} \cap [0, 1]$ itself has measure 0 (Figure 3).

We now understand why each f_n integrates but f itself fails to integrate. Our understanding lies not in the language of the Riemann integral itself but instead in the language of measure.

We also see that the method by which we partition the domain fails for functions that oscillate too quickly and will fail to be defined in spaces outside of $\mathbb{R}$.

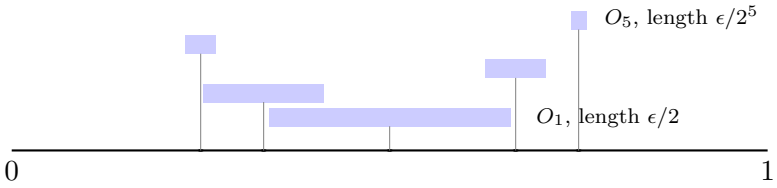


Figure 3: Covering the rationals $r_1, r_2, \ldots$ by intervals of length $\epsilon/2^k$ (stacked vertically for visibility). The total length is at most ϵ, however small ϵ is chosen.

1.4 Lebesgue Savior

Under mild assumptions, integrating with respect to measure gives us

$$\lim_{n\to\infty} \int f_n \, d\mu = 0 = \int f \, d\mu \tag{10}$$

2 Outer Measure

Let $|.|$ denote outer measure.

Definition 8 (Outer Measure)**.** $|A| = \inf \sum_k l(O_k)$ Where inf is taken over all open coverings such that $A \subset \bigcup_k O_k$

2.1 Properties of Outer Measure

Theorem 2.

1. *If* $A \subseteq B$ *then* $|A| \leq |B|$
2. $|[a, b]| = b - a$
3. *If* $A_1, A_2, \ldots$ *satisfy* $\bigcup_k A_k = B$*, then* $|B| \leq \sum_k |A_k|$
4. $|A + t| = |A|$

I will now prove each in order.

Proof: Monotonicity.
For monotonicity (1) first observe that since

$$A \subset B \subset \bigcup_k O_k$$

any open covering of B is an open covering of A.

Let

$$S_a = \left\{ \sum_k l(O_k) : O \text{ is an open cover of } A \right\}$$

Where each O is an open cover of the form $\bigcup_k O_k$.

Similarly let

$$S_b = \left\{ \sum_k l(O_k) : O \text{ is an open cover of } B \right\}$$

$S_b \subset S_a$ since any open covering of B is an open covering of A. It then follows $\inf S_a \leq \inf S_b$.

Giving $|A| \leq |B|$.

Note. Say $B \subset C$ then $\inf C \leq \inf B$; since any lower bound of C is a lower bound of B, making $\inf C$ a lower bound of B, where $\inf B$ is greatest lower bound.

□

Proof: $|[a, b]| = b - a$.
For the second property of outer measure begin by considering the open cover

for $[a, b]$

$$(a - \epsilon, a + \epsilon) \cup (a, b) \cup (b - \epsilon, b + \epsilon)$$

Since $|[a, b]|$ is inf over all coverings we have

$$|[a, b]| \leq 2\epsilon + b - a + 2\epsilon$$

Where ϵ is arbitrary giving

$$|[a, b]| \leq b - a$$

For the other direction consider any open covering of $[a, b]$. I want to show that

$$\sum_k l(O_k) \geq b - a$$

hence giving

$$\inf \sum_k l(O_k) \geq b - a$$

Now $[a, b]$ is a compact interval, by Heine-Borel for every open cover

$$[a, b] \subset \bigcup_k O_k$$

there exists a finite sub-collection such that

$$[a, b] \subset \bigcup_{k=1}^{n} O_k$$

Where

$$\sum_{k=1}^{n} l(O_k) \leq \sum_{k=1}^{\infty} l(O_k)$$

Since length is non negative.

Begin with $n = 1$ intervals. Any interval containing $[a, b]$ ie $[a, b] \subset O_k$ has length $l(O_k) \geq b - a$, since

$$(a, b) \subset [a, b] \subset O_k$$

Suppose the claim holds for n intervals.

Now consider $n + 1$ intervals covering $[a, b]$.

b must be contained in at least one of these intervals, call that interval (c, d). If $c < a$ then $l(c, d) \geq b - a$. So consider $a < c < b < d$. Since (c, d) covers none of $[a, c]$ we can invoke the inductive hypothesis on $[a, c]$ giving

$$\sum_{k=1}^{n} l(O_k) \geq c - a$$

for O_k covering $[a, c]$. Adding this to $l(c, d)$ gives

$$\sum_{k=1}^{n+1} l(O_k) \geq (c - a) + (d - c)$$

$$= d - a$$

$$\geq b - a$$

Completing the proof. □

Proof: Countable Subadditivity.

I want to show

$$|\bigcup_k A_k| \leq \sum_k |A_k|$$

Let

$$A = \bigcup_k A_k$$

If some $|A_k| = \infty$ there is nothing to prove, so assume each $|A_k| < \infty$ and fix $\epsilon > 0$. Since outer measure is inf over all open covers, take for each A_k the open cover $O_k = \bigcup_i O_{ki}$ where

$$\sum_i l(O_{ki}) < |A_k| + \frac{\epsilon}{2^{k+1}}$$

Now since the O_k cover each A_k and $A = \bigcup_k A_k$

$$A \subset \bigcup_k O_k$$

Meaning

$$\begin{aligned} |A| &\leq \sum_k \sum_i l(O_{ki}) \\ &< \sum_k \left(|A_k| + \frac{\epsilon}{2^{k+1}} \right) \\ &= \sum_k |A_k| + \frac{\epsilon}{2} \end{aligned}$$

Which holds for all $\epsilon > 0$ giving

$$|\bigcup_k A_k| \leq \sum_k |A_k|$$

Completing the proof.

□

Proof: Translation Invariance.
I would like to show $|A + t| = |A|$.

First observe that

$$l((a, b)) = l((a + t, b + t))$$

since

$$l((a + t, b + t)) = (b + t) - (a + t) = b - a$$

Now observe that for any open cover of A call it O_A, O_{A+t} covers $A + t$.

Where O_{A+t} is defined as

$$\bigcup_k (O_k + t)$$

for $O_k \in O_A$.

Take any point in $A + t$ call it y. Then $y = x + t$ for some $x \in A$. Where $x \in O_k$ for some $O_k \in O_A$. Then $y \in O_k + t$. Where

$$l(O_k) = l(O_k + t)$$

So translating a cover of A by t gives a cover of $A + t$ with the same total length.

Let

$$S_A = \left\{ \sum_k l(O_k) : O \text{ is an open cover of } A \right\}$$

Where each O is an open cover of the form $\bigcup_k O_k$.

Similarly let

$$S_{A+t} = \left\{ \sum_k l(O_k) : O \text{ is an open cover of } A + t \right\}$$

Meaning since $|A|$ inf over all covers ie $\inf S_A$ and $S_A \subset S_{A+t}$, it follows $|A + t| \leq |A|$.

Now consider the sets $A+t+(-t)$ and $A+t$ where the left side is a translations of the right side. Giving

$$|A + t + (-t)| \leq |A + t| \implies |A| \leq |A + t|$$

Which follows from the above. Completing the proof $|A + t| = |A|$.

□

It seems that outer measure has all the desired properties we would want or need. The problem comes when we want to integrate with respect to measure. We would like to break up the set we are integrating over into disjoint sets whose union is the set itself. Therefore we would additionally want to see.

$$|\bigcup_k A_k| = \sum_k |A_k| \tag{11}$$

This is not much different than our property of sub-additivity. This may seem intuitively true at first, but as we will see in a moment this is not actually true for all sets. Therefore we would like to be explicit about where and when the properties holds to be coherent in our application of measure theory.

2.2 Failure of Outer Measure as a Measure

2.2.1 Equivalence Classes

Definition 9 (Equivalence Relation)**.** x is said to be equivalent to y denoted by $x \sim y$ iff $x - y \in \mathbb{Q}$

To be an equivalence relation it must satisfy the following properties

1. $x \sim x$
2. If $x \sim y$ then $y \sim x$
3. If $x \sim y$ and $y \sim z$ then $x \sim z$.

Proof: Equivalence Relation.
For $x \sim x$, notice for any value x, $x - x = 0$ and $0 \in \mathbb{Q}$.

For $x \sim y \implies y \sim x$, notice that $y - x = -1(x - y)$ where $\mathbb{Q}$ is closed under multiplication.

For $x \sim y$ and $y \sim z$, then $x \sim z$, notice $(x - y) + (y - z) = (x - z)$ where $\mathbb{Q}$ is closed under addition. □

Definition 10 (Equivalence Class on $[0, 1]$)**.** For $x \in [0, 1]$, the equivalence class of x on $[0, 1]$ is the set $\{y \in [0, 1] : y \sim x\} = (\mathbb{Q} + x) \cap [0, 1]$.

Lemma 3. Any two distinct equivalence classes on $[0, 1]$ are disjoint.

Proof: Equivalence Classes Disjoint.
Let A and B be equivalence classes and suppose their intersection is non empty; I will show $A = B$.

Let $z \in A \cap B$. For arbitrary $x \in A$ and $y \in B$, $x - z \in \mathbb{Q}$ and $y - z \in \mathbb{Q}$. Let $x - z = u$ and $y - z = v$.

Solving for x and y gives $x = u + z$ and $y = v + z$. Now subtracting, we get

$$x - y = (u + z) - (v + z) = u - v$$

But $u, v \in \mathbb{Q}$, where $\mathbb{Q}$ is closed under subtraction, giving $x - y \in \mathbb{Q}$, which says $x \sim y$.

So every $x \in A$ is equivalent to an element of B, forcing $x \in B$, making $A \subset B$. Symmetrically $B \subset A$, so $A = B$.

Therefore two equivalence classes are either equal or disjoint. □

2.2.2 Vitali

For the following argument I will use the Axiom of Choice to show that, countable additivity does not necessarily hold for all subsets of $\mathbb{R}$.

Axiom. For any collection of nonempty sets, there exists a function assigning to each set one of its elements.

To begin, using the Axiom of Choice, let a set V contain exactly one element from each equivalence class on $[0, 1]$. The equivalence classes on the interval $[0, 1]$ are of the form $(\mathbb{Q} + x) \cap [0, 1]$. Each class is countable while $[0, 1]$ is not, so there are uncountably many classes, making V uncountable.

Now consider the translations $V + q$ for $q \in [-1, 1] \cap \mathbb{Q}$. Crucially, there are only *countably* many such q, so we may enumerate them $q_1, q_2, \ldots$ and countable additivity applies to this family.

Lemma 4. For any distinct $p, q \in [-1, 1] \cap \mathbb{Q}$, the sets $V + p$ and $V + q$ are disjoint.

Proof: Translated Classes Disjoint.
To see this suppose some element w existed such that

$$w \in (V + p) \cap (V + q)$$

for $p, q \in \mathbb{Q} \cap [-1, 1]$.

Then $w = v + p$ for some $v \in V$ and $w = u + q$ for some $u \in V$. So

$$v - u = q - p$$

making $v - u \in \mathbb{Q}$.

Meaning $v \sim u$, but V only contains one element from each equivalence class, so $v = u$, forcing $p = q$. Hence if $p \neq q$ the translates $V + p$ and $V + q$ are disjoint. □

Now I will show that the union of the translates has bounded, positive outer measure, while a countable sum of equal values can only be 0 or ∞; a contradiction.

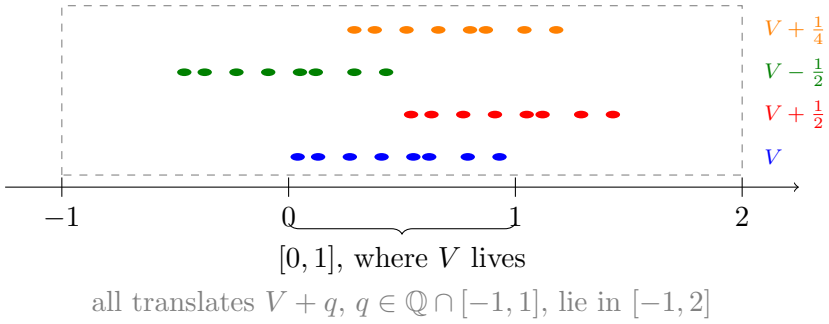


Figure 4: Schematic of the Vitali construction: disjoint rational translates of V that together cover $[0, 1]$ yet stay inside $[-1, 2]$.

Lemma 5 (Vitali Bounded)**.** $[0, 1] \subset \bigcup_q (V + q) \subset [-1, 2]$

Proof: Vitali Bounded.
Take any arbitrary $x \in [0,1]$.

x sits in some equivalence class, and V contains a representative v of that class.

Meaning $x - v = q$ for some $q \in \mathbb{Q}$.

Since both $x, v \in [0,1]$ it implies that $q \in [-1,1] \cap \mathbb{Q}$. Since $\bigcup_q (V+q)$ is a union over all $q \in [-1,1] \cap \mathbb{Q}$, we have

$$[0,1] \subset \bigcup_q (V+q)$$

Now take an arbitrary $v \in V$ and $q \in [-1,1] \cap \mathbb{Q}$. $v \in [0,1]$ and $q \in [-1,1]$ forcing $v + q \in [-1,2]$.

Meaning

$$\bigcup_q (V+q) \subset [-1,2]$$

Completing the proof. □

Now by monotonicity it must be the case that

$$|[0,1]| \leq |\bigcup_q (V+q)| \leq |[-1,2]| \tag{12}$$

Which implies

$$1 \leq |\bigcup_q (V+q)| \leq 3 \tag{13}$$

Since outer measure is translation invariant, $|V+q| = |V|$ for every q, so all of the translates have the same outer measure.

Now suppose it is true for disjoint sets

$$|\bigcup_k A_k| = \sum_k |A_k| \tag{14}$$

then

$$|\bigcup_q (V+q)| = \sum_q |(V+q)| \tag{15}$$

This is a sum of countably infinitely many copies of the single number $|V|$. Suppose that $|V| > 0$; since there are infinitely many $q \in [-1,1] \cap \mathbb{Q}$ it would force $\sum_q |(V+q)| = \infty$.

Suppose then that $|V| = 0$, it would then force $\sum_q |(V+q)| = 0$.

Measure is non negative meaning $|(V+q)| \geq 0$, but either gives a contradiction to

$$1 \leq |\bigcup_q (V+q)| \leq 3 \tag{16}$$

Meaning for the Vitali set (a valid subset of $\mathbb{R}$ given the Axiom of Choice)

$$|\bigcup_k A_k| = \sum_k |A_k| \tag{17}$$

does not hold.

Meaning it is not true in general and therefore we can not assert that

$$|\bigcup_k A_k| = \sum_k |A_k| \tag{18}$$

for the disjoint family $A_k = V + q_k$. As we will see, this means outer measure is not a measure on the collection of all subsets of $\mathbb{R}$.

3 Integration on Measure

First I will define what it means to be a measure and where exactly measures are defined.

Before that I will define a σ-algebra and measurable space.

Definition 11 (σ-algebra)**.** A σ-algebra on a set X is a set of subsets of X satisfying the following three properties.

1. $\emptyset \in S$
2. for any set $E \in S$, $X \setminus E \in S$
3. for any sequence of sets $E_1, E_2, \ldots$ where each $E_k \in S$, $\bigcup_k E_k \in S$

From the properties we can quickly derive $X \in S$. First observe $X \setminus E$ is $\{x \in X : x \notin E\}$. Then from the first two properties $X = X \setminus \emptyset \in S$.

Additionally it follows from De Morgan's laws that for any $E_1, E_2, \ldots$ where each $E_k \in S$, $\bigcap_k E_k \in S$.

$$
\begin{aligned}
E_k \in S &\implies E_k^c \in S \\
&\implies \bigcup_k E_k^c \in S \\
&\implies (\bigcup_k E_k^c)^c \in S \\
&\implies \bigcap_k E_k \in S
\end{aligned}
$$

Where the last line follows from

$$
\begin{aligned}
(\bigcup_k E_k^c) = (\bigcap_k E_k)^c \\
\implies (\bigcup_k E_k^c)^c = ((\bigcap_k E_k)^c)^c \\
\implies (\bigcup_k E_k^c)^c = \bigcap_k E_k
\end{aligned}
$$

Note. Notice that S is defined specifically such that it contains the sets for which we later want to apply countable additivity to.

I will now define a measurable space.

Definition 12 (Measurable Space)**.** A measurable space is a pair (X, S) where X is a set and S is a σ-algebra on X.

Now to define a measure and a measure space.

Definition 13 (Measure Space)**.** Let (X, S) be a measurable space. A function $\mu\colon S \to [0, \infty]$ is a measure if $\mu(\emptyset) = 0$ and $\mu(\bigcup_k E_k) = \sum_k \mu(E_k)$ for every sequence of disjoint sets $E_1, E_2, \ldots \in S$. The triple (X, S, μ) is then called a measure space.

Note. The Lebesgue measure is outer measure restricted to a suitable σ-algebra (the Lebesgue measurable sets), which contains every interval and every countable set but, by the previous section, not the Vitali set. I take its construction as given here.

Now we can define integration with respect to measure and prove our result from earlier. First again I will define partitions and lower sums.

Definition 14 (S-Partitions)**.** Let (X, S, μ) be a measure space, then an S-partition P is defined as

$$P = \{E_1, E_2, \ldots, E_n\} \tag{19}$$

Where each of the E_k are disjoint, $E_k \in S$, and $\bigcup_{k=1}^{n} E_k = X$.

Note. Notice again P is a finite collection of sets, but the sets themselves need not be intervals.

Definition 15 (S-Measurable)**.** Let (X, S) be a measurable space. A function $f\colon X \to [-\infty, \infty]$ is S-measurable if $f^{-1}((a, \infty)) \in S$ for every real number a.

Definition 16 (Lower sums)**.** Let (X, S, μ) be a measure space.

Let f be an S-measurable function $f\colon X \to [0, \infty]$, the lower sums of f for a S-Partition P are defined as

$$L(f, P) = \sum_{k=1}^{n} \mu(E_k) \inf_{E_k} f$$

Now finally integration with respect to measure.

Definition 17 (Integral with Respect to a Measure)**.** Let (X, S, μ) be a measure space and $f\colon X \to [0, \infty]$ an S-measurable function. The integral of f with respect to μ is

$$\int f \, d\mu = \sup_{P} L(f, P) \tag{20}$$

Finally recall our function from earlier

$$f = \lim_{n\to\infty} f_n = \begin{cases} 1: & x \in \mathbb{Q} \cap [0,1] \\ 0: & x \notin \mathbb{Q} \cap [0,1] \end{cases}$$

Consider the S-Partition.

$$\{[0,1] \cap \mathbb{Q}, [0,1] \setminus \mathbb{Q}\}$$

From earlier

$$[0,1] \cap \mathbb{Q}$$

is measure 0, where

$$\inf_{[0,1]\cap\mathbb{Q}} f = 1$$

forcing

$$[0,1] \setminus \mathbb{Q}$$

to be measure 1 since we know $|[0,1]| = 1$ and the two sets union to $[0,1]$.

Where

$$\inf_{[0,1]\setminus\mathbb{Q}} f = 0$$

Integrating with respect to measure then we get the following equation

$$L(f,P) = (0)(1) + (1)(0) = 0 \tag{21}$$

Where from left to right we have, measure 0 times inf 1 plus measure 1 times

inf 0. This shows $\int f\,d\mu \geq 0$.

For the reverse inequality, take any S-partition $\{E_1, \ldots, E_n\}$ of $[0,1]$. If $\mu(E_k) > 0$ then E_k cannot lie inside $\mathbb{Q} \cap [0,1]$, which has measure 0, so E_k contains an irrational and $\inf_{E_k} f = 0$. If $\mu(E_k) = 0$ the term $\mu(E_k) \inf_{E_k} f$ is 0 anyway. Hence every lower sum is 0, and

$$\int f\,d\mu = \sup_P L(f,P) = 0 \tag{22}$$

So the function that Riemann integration could not handle has a perfectly good integral with respect to measure, and it agrees with the limit of the Riemann integrals:

$$\lim_{n\to\infty} \int_0^1 f_n\,dx = 0 = \int f\,d\mu.$$

4 Conclusion

I began with Riemann integration failing to exist at the point wise limit of Riemann integrable functions. To explain why this was the case I used the language of measure. I then showed outer measure which had almost all the properties we wanted failed to be a measure on $\mathbb{R}$. I then defined σ-algebras and measure space in which countable additivity is valid. Finally I showed that our original function could be handled by integrating with respect to measure, where we did not need to use intervals to integrate but only rather the concept of measure.

In general measure gives us the ability to go outside of $\mathbb{R}$ while keeping all of the properties we are looking for at the cost of limiting exactly where our methods hold.